

There are some job opportunities in one company. And there are some candidates for these jobs. Candidate can be hired for the i th job if he has all skills that are needed for this job. There are some distinct skills which can be needed for jobs. Company needs only one person for each job. Moreover each candidate can be hired only for one job and the profit of hiring somebody for the i th job is $profits[i]$. Profit of hiring for the set of jobs equals to sum of profits of each job in set. For the empty set it's 0.

Your task is to calculate a maximum profit after hiring a subset of candidates for a subset of jobs.

Example

For $profits = [4, 10]$, $skillsForJobs = [[true, false], [false, true]]$, and $skillsOfCandidates = [[true, true], [true, false]]$, the output should be

$solution(profits, skillsForJobs, skillsOfCandidates) = 14$;

For $profits = [5]$, $skillsForJobs = [[true]]$, and $skillsOfCandidates = [[true]]$, the output should be

$solution(profits, skillsForJobs, skillsOfCandidates) = 5$.

Input/Output

[execution time limit] 0.5 seconds (cpp)

[input] array.integer profits

Guaranteed constraints:

$1 \leq profits.length \leq 50$,

$0 \leq profits[i] \leq 106$.

[input] array.array.boolean skillsForJobs

If $skillsForJobs[i][j] = true$ then candidate for i th job must have skill with number j .

Guaranteed constraints:

skillsForJobs.length = profits.length,
skillsForJobs[i].length = skillsForJobs[0].length,
 $1 \leq \text{skillsForJobs}[i].\text{length} \leq 50$.

[input] array.array.boolean skillsOfCandidates

If skillsOfCandidates[i][j] = true then ith candidate has skill with number j.

Guaranteed constraints:

$1 \leq \text{skillsOfCandidates}.\text{length} \leq 50$,
 $\text{skillsOfCandidates}[i].\text{length} = \text{skillsForJobs}[0].\text{length}$.

[output] integer